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POLE ASSEMBLY

Abstract

The present invention provides an improved pole assembly. According to a first preferred embodiment, the present invention includes a first pole section which is formed by an inner shell wall surrounding a hollow inner cavity. According to a further preferred embodiment, the inner shell wall is formed as a first column of a given shape, which is formed of galvanized steel or fiberglass reinforced polymer, and which is surrounded by a center fill layer. The center fill layer is preferably formed of SSC concrete which surrounds the inner shell wall. According to a further preferred embodiment, the present invention preferably further includes an outer shell wall which is formed as a second column of a given shape, which surrounds the center fill layer.

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Background/Summary

RELATED APPLICATIONS [0001] The present application is a Continuation-In-Part of U.S. patent application Ser. No. 17/481,438, filed Sep. 22, 2021, which in turn claims priority to U.S. Provisional Application No. 63/082,504 filed Sep. 24, 2020.

FIELD AND BACKGROUND OF THE PRESENT INVENTION

Field of the Present Invention

[0002] The present invention relates generally to an improved pole assembly. More specifically, the present invention relates to a method and apparatus for providing an improved pole assembly which includes inner and outer shell components made of selective materials.

Background of the Invention

[0003] Utility poles are generally single, vertical posts (also referred to as distribution or transmission poles) which are commonly installed at set intervals to support overhead power lines and other public utilities such as electrical cable, fiber optic cable, and related equipment such as transformers and streetlights.

[0004] Most commonly, utility poles are made of pressure treated wood, steel, concrete or composite materials. Composite poles are generally manufactured by centrifugally casting methods, winding and/or pultrusion techniques. Composite poles have an extremely high strength to weight ratio and are highly durable thus lasting longer. Composite poles are also less harmful to the environment. Because of their increased strength, the lengths of composite poles have been pushed to their limits and now commonly extend to sixty feet or higher. At these extended heights, the amount of bending moments and shear force greatly increase. Moreover, because of the reduced stiffness of composite poles, the lateral deflections increase dramatically with increases in length. Increasing the cross-section of composite poles for strength and stiffness requirements can add significantly to the cost of a given structure. For these reasons, current composite pole designs have plateaued in terms of their ability to support weight and to withstand lateral forces. The current pole technologies are either high in cost or require materials which are restrictively heavy in weight, such as concrete, and do not provide the benefits of composite material.

SUMMARY OF THE PRESENT INVENTION

[0005] To address the shortcomings presented in the prior art, the present invention provides an improved utility pole assembly which is light in weight, has increased stiffness, and which is capable of carrying greater applied loads.

[0006] In accordance with preferred embodiments, the present invention includes a first pole section which is preferably formed by an inner and outer shell walls surrounding a hollow inner cavity. According to preferred embodiments, the inner shell wall is preferably formed as a first rectangular, square or circular member.

[0007] According to a further preferred embodiment, the inner shell wall is preferably formed of fiberglass reinforced polymer (FRP) or steel and is preferably surrounded by a center fill layer and an outer shell. The center fill layer (e.g., between the inner and outer shells) is preferably made of high strength concrete (e.g., SSC concrete) or a grout material that is cementitious, resin based or polyurethane.

[0008] According to a further preferred embodiment, the present invention also includes an outer

shell wall which is preferably formed as a second rectangular, square or circular member which surrounds the inner shell and the center fill layer. Preferably, the center fill layer may include reinforcing steel wires which are arranged longitudinally within the center fill layer. According to further preferred embodiments, other forms of interior reinforcements may also be used such as: welded wire reinforcement (WWR), reinforcing steel rebar cage, steel fiber reinforced concrete, FRP bars or FRP laminates. In another embodiment, the interior fill layer may not include additional reinforcement, but depends on the fill material and the two surrounding shells to carry load.

[0009] Other goals and advantages of the invention will be further appreciated and understood when considered in conjunction with the following description and accompanying drawings. While the following description may contain specific details describing particular embodiments of the invention, this should not be construed as limitations to the scope of the invention but rather as an exemplification of preferable embodiments. For each aspect of the invention, many variations are possible as suggested herein that are known to those of ordinary skill in the art. A variety of changes and modifications can be made within the scope of the invention without departing from the spirit thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Elements in the figures have not necessarily been drawn to scale in order to enhance their clarity and to improve the understanding of the various elements and embodiments of the invention. Furthermore, elements that are known to be common and well understood to those in the industry are not depicted in order to provide a clear view of the various embodiments of the invention. Thus, it should be understood that the drawings are generalized in form in the interest of clarity and conciseness.

[0011] FIG. 1 is a side cut-away view of an exemplary pole section in accordance with the present invention.

[0012] FIG. 2A is a cross-sectional view of the exemplary pole shown in FIG. 1 cut along the line A-A.

[0013] FIG. 2B is a cross-sectional view of the exemplary pole shown in FIG. 1 cut along the line B-B.

[0014] FIG. 3 is a side cut-away view of an exemplary pole section in accordance with a first alternative preferred embodiment.

[0015] FIG. 4A is a cross-sectional view of the exemplary pole section shown in FIG. 3 cut along the line C-C.

[0016] FIG. 4B is a cross-sectional view of the exemplary pole section shown in FIG. 3 cut along the line D-D.

[0017] FIG. 5 is a side cut-away view of a pole section in accordance with a further alternative preferred embodiment.

[0018] FIG. 6 is a cross-sectional view of the exemplary pole section cut along the line E-E shown in FIG. 5.

[0019] FIG. 7A is an exemplary cross-sectional view of an exemplary pole section having circular inner and outer tube walls.

[0020] FIG. 7B is an exemplary cross-sectional view of an exemplary pole section having octagonal inner and outer tube walls.

[0021] FIG. 8 is a perspective view of an exemplary pole produced in accordance with the present invention.

[0022] FIG. 9 is a side cut-away view of a further alternative preferred embodiment of the present

invention.

[0023] FIG. **10A** is a side cut-away view of a further alternative preferred embodiment of the present invention.

[0024] FIG. **10B** is a cross-sectional view of the exemplary pole shown in FIG. **10A** cut along the line F-F.

[0025] FIGS. **11A-11C** are cross-sectional views of a pole assembly in accordance with exemplary alternative embodiments of the present invention.

[0026] FIG. **12** is a side cut-away view of a pole assembly in accordance with a further alternative preferred embodiment of the present invention.

[0027] FIG. **13** is a cross-sectional view of the exemplary pole shown in FIG. **13** cut along the line G-G.

[0028] FIG. **14** illustrates a set of cross-sectional views (a)-(e) illustrating alternative exemplary embodiments of the present invention.

[0029] FIG. **15** is an exemplary cross-section of an exemplary pole in accordance with a further preferred embodiment.

[0030] FIG. **16** is a cross-section of an exemplary pole in accordance with a further preferred alternative embodiment which includes interior ribs.

[0031] FIG. **17** is an exemplary cross-section of the exemplary pole shown in FIG. **16** with multiple interior ribs.

[0032] FIG. **18** is a cross-section of an exemplary pole with an alternative interior rib design.

[0033] FIG. **19** is an exemplary cross-section of the exemplary pole shown in FIG. **18** with multiple interior ribs.

[0034] FIG. **20** shows a further alternative embodiment which includes an interior rib which is attached to the outer tube, and which encloses a vertical steel rod.

[0035] FIG. **21** is a cross-section of an exemplary pole in accordance with a further preferred alternative embodiment which includes patterned interior surfaces of tube walls.

[0036] FIG. **22** is a cross-section of a pole assembly in accordance with a further alternative preferred embodiment of the present invention.

[0037] FIG. **23** is a cross-sectional view of the exemplary pole shown in FIG. **22** cut along the line H-H.

[0038] FIG. **24** is a cross-sectional view of a pole assembly in accordance with a further alternative preferred embodiment of the present invention.

[0039] FIG. **25** is a cross-sectional view of the exemplary pole shown in FIG. **24** cut along the line J-J.

[0040] FIG. **26** is a side cut-way view of an exemplary pole with cross-sectional views illustrating a partial outer sleeve application.

[0041] FIG. **27** is a set of cross-sectional views illustrating selected exemplary embodiments of the present invention.

[0042] FIG. **28** illustrates an exemplary method for creating pole structures in accordance with the present invention.

[0043] FIG. **29** is a side cut-away view of a further alternative exemplary pole.

[0044] FIG. **30** is a side cut-away view of a further alternative exemplary FRP-Concrete-FRP hybrid pole.

[0045] FIG. **31** is a side cut-away view of a further alternative exemplary FRP-Concrete-Steel hybrid pole.

[0046] FIG. **32** is a cross-sectional view of the pole shown in FIG. **31** cut along line M-M.

[0047] FIG. **33** is a cross-sectional view of the pole shown in FIG. **31** cut along line N-N.

[0048] FIG. **34** is a cross-sectional view of an exemplary pole illustrating a further alternative embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0049] Aspects of the present invention will be explained with reference to exemplary embodiments and examples which are illustrated in the accompanying drawings. These descriptions, embodiments and figures are not to be taken as limiting the scope of the claims.

[0050] Where the specification describes advantages of an embodiment or limitations of other prior art, the applicant does not intend to disclaim or disavow any potential embodiments covered by the appended claims unless the applicant specifically states that it is “hereby disclaiming or disavowing” potential claim scope. Likewise, the term “embodiments” does not require that all embodiments of the invention include any discussed feature or advantage, nor that it does not incorporate aspects of the prior art which are sub-optimal or disadvantageous.

[0051] Additionally, any examples or illustrations given herein are not to be regarded in any way as restrictions on, limits to, or express definitions of, any term or terms with which they are utilized. Instead, these examples or illustrations are to be regarded as illustrative only. Those of ordinary skill in the art will appreciate that any term or terms with which these examples or illustrations are utilized will encompass other embodiments which may or may not be given therewith or elsewhere in the specification and all such embodiments are intended to be included within the scope of that term or terms.

[0052] As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Additionally, the word “may” is used in a permissive sense (i.e., meaning “having the potential to”), rather than the mandatory sense (i.e., meaning “must”). Further, it should also be understood that throughout this disclosure, unless logically required to be otherwise, where a process or method is shown or described, the steps of the method may be performed in any order (i.e., repetitively, iteratively or simultaneously) and selected steps may be omitted. It will be further understood that the terms “comprises,” “comprising,” “includes” and/or “including,” when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0053] Further, the word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Accordingly, any embodiment described herein as “exemplary” is not to be construed as preferred over other embodiments. Additionally, well-known elements of the embodiments will not be described in detail or will be omitted so as not to obscure relevant details.

[0054] With reference now to FIGS. **1-33**, the present invention teaches an improved pole assembly. As discussed below, the improved pole assembly of the present invention preferably includes a vertical pole structure incorporating one or more pole sections built in accordance with aspects of the present invention. According to alternative preferred embodiments, a pole assembly/pole of the present invention may be formed of a single pole section as discussed below. Alternatively, an improved pole assembly/pole may include multiple different pole sections combined from different embodiments of the present invention.

[0055] With reference now to FIG. **1**, a side cut-away view of an exemplary utility pole section **10** in accordance with the present invention shall now be discussed. As shown in FIG. **1**, the exemplary pole section **10** is preferably formed of a hollow inner cavity **12** which is enclosed within a multi-layer wall **14**. As shown, the multi-layer wall **14** preferably includes an outer shell **18**, a center fill layer **20** and an inner shell **16**. The inner and outer shells **16, 18** are preferably formed of fiberglass reinforced polymer (FRP). Alternatively, the inner and outer shells **16, 18** may be formed of any steel, fiberglass or any composite or other structural material. When formed of FRP, the inner and outer shells **16, 18** may preferably be formed of a single FRP layer or as FRP tubing. According to a preferred embodiment, the center fill layer **20** is preferably formed of a high strength concrete filler material or the like. Preferably, the center fill layer **20** is formed from self-consolidating concrete (SSC) or the like. Alternatively, the center fill layer **20** may be formed of other pumpable material such as high strength concrete, polyurethane, or other foam, plastic or

similar material. Additionally, the center fill layer **20** may be filled with different filler materials at various heights of the pole (e.g., such as the lower portion **11** and the upper portion **13** and as discussed further below). As shown, the center fill layer **20** may additionally support and enclose vertically extending reinforcing wires **22** or the like.

[0056] According to preferred embodiments, the exemplary pole section **10** is preferably formed as a rectangular column (or alternatively a square or circular column as discussed further below). Accordingly, the inner shell wall **16** preferably is formed as a first rectangular column, and the outer shell wall **18** is preferably formed as a second rectangular column surrounding the first rectangular column. With reference now to FIGS. 2A and 2B, two cross-sectional views of the exemplary pole section **10** shown in FIG. 1 are provided. More specifically, FIG. 2A is a cross-sectional view of the exemplary pole shown in FIG. 1 cut along the line A-A. FIG. 2B is a cross-sectional view of the exemplary pole shown in FIG. 1 cut along the line B-B. As shown in FIG. 2A, the reinforcing wires within a lower portion **11** of the pole section may preferably be formed of two pairs of reinforcing wires **23** which run vertically within the wall section **14**. As shown in FIG. 2B, the reinforcing wires within an upper portion **13** of the pole section may preferably be formed of a single pair of reinforcing wires **25** which run vertically within the wall section **14**. According to alternative preferred embodiments, each section of the assembled pole **10** may include different numbers and layouts of reinforcing wires **22**. Further, each section may include different types of reinforcing wires or bars and may also use other reinforcing materials which each may vary depending on the respective height of the pole section being reinforced.

[0057] According to preferred embodiments, the reinforcing wires **22**, **23**, **25** may preferably be formed of high strength steel and may preferably be pre-stressed. Alternatively, the reinforcing wires **22**, **23**, **25** may be formed from other metals or fiber reinforced polymer (FRP) material. Still further, in the present example and for each embodiment discussed in the present application, additional and/or alternative reinforcing materials may be used such as welded wire reinforcement (WWR), welded wire sheets/rolls, welded wire fabric (WWF), welded wire mesh (WWM), rebar and/or shaped FRP composites (e.g., FRP bars, FRP mats, FRP cruciform, FRP laminates) and the like, alone or in combination with other materials. As used throughout the present application, each of these materials may be used in place of or in combination with any other type of reinforcement materials without limitation.

[0058] As discussed further below, the geometry of each pole cross-section may be different to include any of a variety of geometric shapes. Such shapes may include triangular, circular, semi-circular, square, and any polygonal shapes (e.g., pentagon, hexagon, octagon, nonagon, decagon, etc.). Additionally, the walls **14** of each pole section may be straight (as shown) or may be tapered as discussed further below. The manufacturing of each pole section may preferably be accomplished using pultrusion methods. Alternatively, each pole section may be centrifugally cast or may be molded/assembled via hand lay-up.

[0059] FIG. 3 is a side cut-away view of an assembled pair of pole sections **27** in accordance with a first alternative preferred embodiment of the present invention. As shown in FIG. 3, the pole sections **27** preferably include a lower multi-layer wall section **17** which includes an inner shell wall **16** and an outer shell wall **18** which extends vertically and terminates at a top surface **15**. As further shown, the inner shell wall **16** preferably may continue to an upper portion **19** which extends above the top surface **15** of the lower multi-layer wall section **17**. As shown in a further illustration provided in FIG. 3, the lower multi-layer wall section **17** may extend to a length of 35 feet with an exterior width or diameter of 10 inches. As further shown, the lower wall section **17** may be installed with 7 feet of its length below grade and 28 feet above grade. The upper portion **19** may for example extend 10 feet beyond the top surface **15** of the lower multi-layer wall section **17**. As further shown, the upper portion **19** may have an outer diameter of 7 inches. FIG. 4 provides a cross-sectional view of the exemplary pole shown in FIG. 3 cut along the line C-C including one pair of reinforcing wires/bars **25**. FIG. 4B is a cross-sectional view of the exemplary pole section

shown in FIG. 3 cut along the line D-D. Still further, FIG. 8 provides a perspective view of an exemplary pole **71** which illustrates exemplary pumping ports **72** to allow for the injection of SSC or other materials into the center wall **20** of the pole section.

[0060] With reference now to FIG. 5, another alternative preferred embodiment of the present invention shall be discussed. As shown, an alternative preferred embodiment includes a pole **29** which includes an encasing base **26** which preferably secures the pole section **29** to the ground. As shown, the pole **29** preferably includes a multi-layer wall section **21** which includes an inner wall shell **16** and an outer shell wall **18** which each extend vertically, and which intersect with a top surface **15**. Preferably, the inner shell wall **16** may continue as an upper portion **23** which extends above the top surface **15** of the multi-layer wall section **21**. As further shown, the outer shell wall **18** preferably includes a lower wall segment **28** which vertically extends to the bottom of the encasing base **26**. As shown, the reinforcing wires **30** within the lower multi-layer wall section **21** preferably extend down through the encasing base **26**. The inner shell **16** may not extend all the way to the encasing base **26** and may instead terminate at a specified distance (e.g., 5', 3') from bottom of the base **26**.

[0061] FIG. 6 provides a cross-sectional view of the exemplary pole **29** shown in FIG. 5 cut along the line E-E. As shown, the pole **29** may preferably include multiple pairs of reinforcing wires **36-42** which are located on the corners of the pole **29**. As further shown, the pole **29** of the present invention may preferably further include reinforcing wires which run through the center fill layer of each face of the pole **29**.

[0062] As discussed above, the exemplary pole sections of the present invention may preferably be formed as a rectangular column with the shell walls **16, 18** each formed as separate rectangular columns. According to alternative preferred embodiments, the exemplary pole sections of the present invention may alternatively be formed in any of a variety of geometric shapes (e.g., triangular, circular, semi-circular, square, and any polygonal shape). For example, as shown in FIG. 7A, an exemplary pole **52** may be formed with a circular shaped multi-layer wall **53** which includes a circular shaped inner FRP tube **56** and outer FRP tube **58** which create and surround a circularly shaped center fill layer **60**. According to a preferred embodiment, reinforcing wires **62** (as discussed above) may preferably be equally spaced within the multi-layer wall **53** or can be of any other desired layout. The inner and outer tubes may also be of structural steel or other materials.

[0063] As a further example, FIG. 7B illustrates an exemplary pole **64** which may be formed with an octagonally shaped multi-layer wall **69** which includes an octagonally shaped inner FRP tube **65** and outer FRP tube **67** which create and surround an octagonally shaped center fill layer **61**.

According to a preferred embodiment, reinforcing wires **68** (as discussed above) may preferably be inserted and spaced adjacent to each intersection point **70** between each of the faces of the outer FRP tube **67**.

[0064] With reference now to FIG. 9, according to a further alternative preferred embodiment of the present invention, the center fill layer(s) of the present invention may be formed of different materials at different levels within the pole sections of the present invention. For example, as shown in FIG. 9, an example pole wall **75** may include a lower portion **74** which includes a center fill layer filled with a first material such as concrete/SSC. Additionally, the same pole wall **75** may also include an upper portion **76** which includes a center fill layer filled with a second materials such as polyurethane, foam, foam resin or the like.

[0065] With reference now to FIGS. 10A and 10B, an additional alternative preferred embodiment of the present invention shall now be discussed. FIG. 10A illustrates an example pole assembly **78** which includes a multi-layer wall **80** which includes an inner FRP tube **82** and an outer FRP tube **84** which enclose a center fill layer **85**. FIG. 10B provides a cross-sectional view of the exemplary pole shown in FIG. 10A cut along the line F-F. According to a preferred embodiment, the center fill layer **85** is preferably filled with concrete/SSC or other materials as discussed above. Additionally, the center fill layer **85** preferably includes vertically oriented reinforcing wires or bars **86**

(providing longitudinal reinforcement) which may preferably be spaced evenly around the interior of the center fill layer **85**. According to a further preferred embodiment, the example pole assembly preferably also includes laterally extending reinforcing wire **88** (providing transverse reinforcement) which are preferably vertically spaced within the center fill layer (i.e., spaced at different heights within the center fill layer **85**).

[0066] With reference now to FIG. **11A**, an enlarged cross-sectional view of an exemplary pole assembly **90** is shown. As shown, an inner FRP tube **92** and an outer FRP tube **94** may enclose a center fill layer **95**. Additionally, the center fill layer **95** may support vertically reinforcing FRP bars **98** (or other reinforcing elements such as FRP cruciform elements or other stiffeners as discussed herein). As shown in FIG. **11A**, the vertically reinforcing bars **98** (or other reinforcing elements) may vary in size and their widths may be close to the width of the center fill layer **95** or they may be monolithically formed with the inner and outer FRP tubes **92**, **94**. Where the reinforcing elements **98** approach or match the width of the center fill layer **95**, the pole may include multiple pumping holes to inject concrete/SSC, foam or the like. According to a preferred embodiment, these may be positioned along fill gaps **96** created between the reinforcing elements **98**. According to a preferred embodiment, the reinforcing elements **98** may be FRP bars which may have a diameter of 1½ inches and the FRP tubes **92**, **94** may be approximately 3/16" or 3/8" in thickness. As shown in FIGS. **11B** and **11C**, FRP cruciform elements **97** and/or other stiffeners may be used within the center fill layer to provide targeted support for centering the outer and inner tubes.

[0067] With reference now to FIG. **12**, an additional exemplary pole assembly **100** is shown. As shown, the pole assembly **100** includes a multi-layer wall **101** which includes an inner FRP tube **102** and an outer FRP tube **104** enclosing a center fill layer **105**. Additionally, the center fill layer **105** may support vertically reinforcing wire **106** as discussed above. Additionally, the exemplary pole assembly **100** may additionally include transversely aligned, wire reinforcements **108** which may be formed as hoops which may run at different heights within the interior of center fill layer **105**. Preferably, the hoops **108** may vary in thickness/strength depending on their heights within the final assembled pole **100**. As shown, the multi-layer wall **101** may be tapered to form a variety of cone and pyramid type shapes. FIG. **13** is a cross-sectional view of the exemplary pole shown in FIG. **12** cut along the line G-G

[0068] FIG. **14** is a set of cross-sectional views (a)-(e) illustrating selected exemplary embodiments of the present invention as discussed above. As shown in a first exemplary embodiment (a), the pole **110** may include a pair of tube walls **111**, **113** which extend together the full length of the pole **110**. According to alternative preferred embodiments, the pole **110** may include reinforcement which may extend to varying lengths within the walls.

[0069] In a further illustrated embodiment (b), the tube walls **115**, **117** may extend the full length of the pole **112** but different fill materials may be used within different sections of the pole **112**. Accordingly, an upper section of the pole **119** may be filled with polyurethane (or other materials) and a lower section **121** may be filled with SSC (or other materials).

[0070] In a further illustrated embodiment (c), the pole **114** may include an outside FRP tube **123** which may preferably not run the full length of the pole **114**. Instead, the pole **114** may preferably include an inner FRP tube **125** which may preferably run the full length of the pole **114** and which may be enclosed within the outside FRP tube **123** for only a selected length.

[0071] In a further illustrated embodiment (d), the pole **116** may include an outside FRP tube **127** which may preferably not run the full length of the pole **116** and which may terminate in a securing base **131** which may be filled with fill materials (e.g., SSC, polyurethane or the like). As shown, the pole **116** may preferably include an inner FRP tube **129** which may be enclosed within the outside FRP tube **127** for only a selected length.

[0072] In a further illustrated embodiment (e), the pole **118** may include a lower pole section **133** and an upper pole section **135**. As shown, the lower pole section **133** may include an outside FRP

tube **137** and an inner FRP tube **139** which together enclose filler and/or reinforcement materials. As shown, the upper pole section **135** may also include an outer FRP tube **141** and an inner FRP tube **143** which may also enclose filler and/or reinforcement materials. As further shown, the lower pole section **133** preferably encloses at least a section of the upper pole section **135**.

[0073] With reference now to FIG. **15**, a cross-section **120** of an exemplary pole in accordance with a further preferred embodiment is provided. As shown, the exemplary pole section **120** may preferably include an inner FRP tube **128** and an outer FRP tube **130** which together enclose an interior filler section **134**. The filler section **134** may preferably include reinforcement materials **136** (e.g., welded wire reinforcement (WWR), mesh, welded wire sheets and the like). As further shown in FIG. **15**, a first portion **136** of the filler section **134** may also include a first set of filler material (e.g., SCC or the like). Additionally, a second portion **132** of the filler section **134** may further include a second set of filler material (e.g., polyurethane foam or the like). This may specifically be used in the embodiments shown in FIGS. **14B** and **14E** where different filler materials may be used in the upper and lower sections. Additional sections and filler materials may preferably be added without limitation. As noted above, any other type of cross-sectional shape (or structural material) may also be used without limitation.

[0074] With reference now to FIG. **16**, a cross-section **138** of an exemplary pole in accordance with a further alternative embodiment is provided. As shown, the present invention may preferably further include interior ribs **140** within the inner and outer FRP walls **128**, **130**. Preferably, the interior ribs **140** may be built integrally with the inner FRP tube **128**. As shown in FIG. **17**, multiple interior ribs **142-156** may preferably be spaced (e.g., ~4") throughout the pole interior.

[0075] Referring now to FIG. **18**, a cross-section **145** of an exemplary pole showing an alternative interior rib **158** is provided. As shown, the interior rib **158** of the present invention may preferably be formed in two pieces **147**, **149** which may interlock or otherwise mechanically engaged to form a single rib **158**. As shown, the first rib element **147** may preferably be integrally formed with the outer FRP wall **130** and the second rib element **149** may be formed with the interior FRP wall **128**. FIG. **19** is an exemplary cross-section of the exemplary pole shown in FIG. **18** with multiple interior ribs **160-172**. These may preferably be evenly spaced (e.g., ~4" apart or the like) throughout the pole interior.

[0076] Referring now to FIG. **20**, a cross-section **175** of an exemplary pole showing an alternative exemplary rib **176** is provided. As shown, the alternative interior rib **176** may preferably be built integrally with (or attached to) the outer FRP outer tube **130** and may extend laterally towards the inner FRP tube **128**. As shown, the interior rib **176** may additionally fully or partially extend around vertically extending steel bars **178**. In this design, the interior rib(s) **176** may preferably function to position the vertical steel bars and hold them in place. Additionally, the interior rib(s) **176** may provide spacing/alignment for the FRP tubes **128**, **130** during construction of the pole and pumping of the filler material. As shown, the interior rib **176** may not extend fully between the FRP tubes **128**, **130**. However, in alternative embodiments, the interior rib **176** may be attached between, touch and/or be frictionally fit between the two FRP tubes **128**, **130**. Additionally, multiple ribs may preferably be used and repeated every few inches (e.g., ~2" to 4") to contain multiple vertically aligned steel bars.

[0077] FIG. **21** is a cross-section **180** of an exemplary pole in accordance with a further preferred alternative embodiment which includes interior and exterior FRP shells/walls **128**, **130** having patterned interior surfaces **182**, **184**. The patterned interior surfaces **182**, **184** may preferably include contours, deformations, ribs, projections and the like without limitation.

[0078] With reference now to FIGS. **22** and **23**, a further alternative embodiment of the present invention shall now be discussed. FIG. **22** provides a cross-sectional view of an exemplary pole assembly **186** which includes an outer shell/tube **190** which is formed of FRP and an inner shell/tube **188** which is preferably formed of steel. Preferably, a central layer **189** is formed between the two shells/tubes **188**, **190**. Preferably, the central layer **189** is formed of concrete or

high strength grout to form one monolithic, composite section that works together in supporting applied loads. As further shown, steel reinforcement (e.g., rebar) **191** may further be used (if required by design) between the two shells/tubes **188, 190**. According to alternative embodiments, the thickness of the concrete or grout may vary, and may for example be between 1"-2" (or any other desired thickness). FIG. **23** is a cross-sectional view of the exemplary pole shown in FIG. **22** cut along the line H-H.

[0079] With reference now to FIGS. **24-26**, a further alternative preferred embodiment **193** is shown which includes an outer shell/tube **197** which is formed of FRP. The outer shell/tube **197** may be spun with the pre-stressed concrete pole **196** during production to form a monolithic sleeve over the concrete pole **196**. Preferably, the concrete pole **196** may act as one composite section to support applied loads. As further shown, steel reinforcement **198** (e.g., pre-stressed strand or wire) may be used in the concrete pole **196**. As further shown, supporting hoops **199** may further be inserted and used within the concrete pole **196**. FIG. **25** is a cross-sectional view of the exemplary pole shown in FIG. **24** cut along the line J-J.

[0080] According to a preferred embodiment, the exemplary pole shown in FIGS. **24-26** may preferably be manufactured in a steel mold or the like. In a preferred manufacturing process, FRP material may be placed within a steel mold and pushed to a desired length. Thereafter, a steel cage may then be inserted in the steel mold and through the FRP tube. The longitudinal steel wires may preferably be high strength reinforcing steel or may otherwise be strengthened via pre-stressing or a similar process. Following this step, high strength concrete may be pumped into the steel mold and within the embedded FRP tube (shell). Thereafter, the FRP tube and the fresh concrete may be taken to a spinner and spun for a prescribed number of minutes. The result is preferably a pre-stressed, reinforced concrete pole with an FRP outer shell.

[0081] Referring now to FIG. **26**, a side cut-away view of a further alternative exemplary pole **200** with cross-sectional views is provided. As shown, the FRP outer shell **204** may extend over various sections and lengths of the inner concrete **202**. As shown, the FRP outer shell **204** does not extend to cross-section K-K. In this way, the sleeve lengths of the FRP outer shell **204** may be selectively applied to protect and support targeted pole sections, while leaving other sections uncovered.

Embodiment 1—FRP-CONC-FRP Hybrid

[0082] Referring now to FIG. **29**, a side cut-away view of a further alternative exemplary pole **309** is provided illustrating an exemplary FRP-Concrete-FRP hybrid pole. As shown, the FRP outer shell **312** may extend over various sections and lengths of the inner concrete layer **310**. The inner concrete (or grout or foam) layer **310** preferably entirely surrounds and supports a central FRP inner shell **311** which is preferably hollow. As further shown, one or more lateral supports/spacers **313** may preferably be extended through the concrete layer **310** and FRP outer shell **312** for alignment purposes during construction of the pole.

[0083] Referring now to FIG. **30**, a side cut-away view of a further alternative exemplary FRP-Concrete-FRP hybrid pole **314** is provided. As shown, the FRP outer shell **318** may extend over various sections and lengths of the inner concrete layer **315**. The inner concrete layer **315** is applied around a central FRP inner shell **316** (as discussed further below) which is preferably hollow. As further shown, groups of lateral supports/spacers **317** may preferably be extended through the concrete layer **315** and FRP outer shell **318**. According to preferred embodiments, the thickness of the outer FRP shell **318** may preferably be approximately $\frac{3}{8}$ " at the extreme tip and butt ends of the pole and may otherwise be approximately $\frac{1}{4}$ " average thickness at other points.

[0084] The dimensions of each example hybrid pole **309, 314** are shown in respective FIGS. **29** and **30**. These and other preferable features are further listed in Table I below.

TABLE-US-00001 TABLE I Embodiment 1-FRP-CONC-FRP Hybrid Outer Shell Centrifugally Cast FRP Tube. May include: 5% black pigment and BYK additive. one layer of polyester veil for outermost surface. two layer glass layup 8000 on outside and 1810 in inside. no paint coating Fill Layer Concrete, high strength grout, or polyurethane Inner Shell Filament Wound Inner FRP Tube

Dimensions a) $\approx 14.96''$ Butt/ $\approx 8.66''$ Tip (FIG. 29) b) $\approx 16.36''$ Butt/ $\approx 8.66''$ Tip (FIG. 30) Taper $\approx .14$ in./ft. Concrete a) $\approx 1.5''$ at Tip to $\approx 2.0''$ at Butt. Thickness b) $\approx 1.0''$ at Tip to $\approx 1.4''$ at Butt.

Embodiment 2—FRP-CONC-STEEL Hybrid

[0085] Referring now to FIG. 31, a side cut-away view of a further alternative exemplary pole 319 is provided illustrating an exemplary FRP-Concrete-Steel hybrid pole. As shown, the FRP outer shell 322 may extend over various sections and lengths of the inner concrete layer 321. The inner concrete layer 321 is applied around a central steel pole 320 which is preferably hollow. FIG. 32 provides a cross-sectional view of the hybrid pole 319 cut along line M-M. FIG. 33 provides a cross-sectional view of the hybrid pole 319 cut along line N-N. As shown, weldments 325 may be attached to the outer surface of the central steel pole as discussed further herein to create a uniform annular space for pumped concrete or the like.

[0086] According to preferred embodiments, the thickness of the outer FRP shell 322 may preferably be approximately $\frac{3}{8}''$ at the extreme tip and butt ends of the pole and may otherwise be approximately $\frac{1}{4}''$ average thickness at other points. The inner and outer tubes 320 and 322 may be tapered and may accommodate different concrete thicknesses (e.g., 1", 1.5" and 2"). These and other exemplary features are listed in Table II below.

TABLE-US-00002 TABLE II Embodiment 2-FRP-CONC-STEEL Hybrid Outer Shell

Centrifugally Cast FRP Tube Fill Layer Concrete, high strength grout, for polyurethane Inner Shell Hot Dipped Galvanized Steel Inner Tube Dimensions $\approx 15.6''$ Butt, $\approx 11''$ Tip Taper $\approx .14$ in./ft.

Concrete a) Stepped 1) $\approx 1.0''$; 2) $\approx 1.5''$; 3) $\approx 2.0''$ Thickness b) Uniform taper $\approx 1.0''$, 1.25", or 1.5"

[0087] With reference now to FIG. 28, an exemplary method 300 for creating an exemplary FRP-Concrete-Steel hybrid pole 319 as shown in FIGS. 31-34 shall now be discussed. At a first step 301, outer spacing elements are preferably first attached to the steel pole outer surface. The spacer elements may for example be vertically aligned weldments 325 (shown in FIGS. 32-33), attached railings and the like. The spacer elements preferably are evenly spaced around the steel pole outer surface so as to provide uniform spacing around the steel pole to create a uniform annular space for later pumped concrete as discussed further below.

[0088] At a next step 302, the steel tube with the attached spacers is preferably inserted within an extended outer FRP shell. At a next step 303, one or more rubber gasket spacers are preferably applied to each end spacing element. Additionally, one or more rubber gasket spacers may further be applied to the end surfaces of each of the top and bottom sides of the inserted steel tube. At a next step 305, silicon is preferably applied to contact surfaces of a top filling cap and a bottom filling cap. One or more of the filling caps may preferably include input nozzles for receiving pumped concrete. According to a preferred embodiment, the contact surfaces may preferably include each point of contact between the steel pole and the surfaces of the top and bottom filling caps as they are secured to respective ends of the assembled hybrid pole.

[0089] At a next step 306, the top and bottom filling caps are preferably attached to the outer FRP shell thus sealing the internal cavity of the outer FRP shell. At a next step 307, the pumping of a pre-mixed concrete mixture (or other filler material) is preferable initiated through one or both of the top/bottom filling caps. At a next step 308, the system preferably initiates movement/vibration which is applied to the sealed outer FRP shell. According to a preferred embodiment, the movement/vibration may preferably be provided by a vibrating stand/cradle supporting the sealed FRP shell. Alternatively, vibrating elements may be secured to the outside surface of the outer FRP shell. In this way, the concrete may preferably be pumped within the outer FRP shell in an amount sufficient to fill the uniform spacing around the steel pole to create a uniform concrete layer once cured.

[0090] The present invention has been disclosed above with reference to several exemplary embodiments which are not intended to be limiting. For example, the shapes and sizes of each example pole discussed herein may be formed in any of a variety of other shapes and sizes. FIG. 27 for example illustrates a group of example pole cross-sections 206-214 which may be used.

[0091] Further, each feature of each exemplary pole of the present invention may be combined with, substituted for, and mixed and matched with every other feature disclosed herein to form hybrid and composite poles without limitation. For example, the inner concrete layers discussed above with respect to FIGS. 29-34 may also include each of the concrete reinforcement features discussed with respect to FIGS. 1-26 without limitation. Additionally, polyurethane and other filling materials discussed herein may be substituted for the concrete layers discussed herein such as with respect to the embodiments shown in FIGS. 29-34. Likewise, the centering weldments attached to a central steel pipe may be substituted for the FRP inner shell walls discussed with respect to FIGS. 1-26 without limitation.

[0092] As further shown in FIG. 34, weldments 327-331 for use with the present invention may be formed in a variety of supporting shapes which may run along the pole length. These weldments 327-331 may be attached and positioned to provide alignment between the FRP tube 334 and the inner steel tube 332 during construction. Additionally, they may be attached as shear connectors tying the inner steel tube 332 to the concrete layers 333.

[0093] The scope of the present invention should be determined purely by the terms of the appended claims and their legal equivalents.

Claims

1. A pole assembly, wherein the pole assembly comprises: a hollow inner cavity; an inner shell wall, wherein the inner shell wall comprises a first column; wherein the inner shell wall surrounds the hollow inner cavity; wherein the inner shell wall comprises a steel or FRP pole; a center fill layer, wherein the center fill layer surrounds the inner shell wall; wherein the center fill layer comprises SSC concrete; and an outer shell wall, wherein the outer shell wall comprises a second column; wherein the outer shell wall comprises fiberglass reinforced polymer; wherein the outer shell wall surrounds the inner shell wall and the center fill layer.
2. The pole assembly of claim 1, wherein the inner shell wall comprises a galvanized steel pole.
3. The pole assembly of claim 2, wherein the galvanized steel pole comprises attached spacer elements.
4. The pole assembly of claim 3, wherein the spacer elements are evenly spaced around the outside circumference of the galvanized steel pole.
5. The pole assembly of claim 3, wherein at least one spacer element comprises a weldment.
6. The pole assembly of claim 5, wherein the spacer elements form a uniform spacing around the steel pole when the steel pole is inserted within the outer shell wall.
7. The pole assembly of claim 6, wherein the center fill layer is formed of concrete.
8. The pole assembly of claim 7, wherein the center fill layer comprises SSC concrete.
9. The pole assembly of claim 8, wherein the center fill layer comprises a first reinforcement wire comprising high strength steel at a first lower level; wherein the center fill layer comprises a second reinforcement wire comprising fiberglass reinforced polymer at a second upper level.
10. The pole assembly of claim 8, wherein the center fill layer comprises SSC at the first lower portion; wherein the center fill layer comprises polyurethane at the second upper portion.
11. The pole assembly of claim 10, wherein the center fill layer comprises a plurality of laterally extending reinforcing wires which are vertically spaced within the center fill layer.
12. The pole assembly of claim 11, wherein the laterally extending reinforcing wires are formed as hoops.
13. The pole assembly of claim 12, wherein the center fill layer comprises a plurality of interior ribs.
14. The pole assembly of claim 13, wherein at least one interior rib comprises two interlocking pieces which are mechanically engaged to form a single rib.
15. The pole assembly of claim 14, wherein at least a portion of the single rib is attached to the

outer FRP outer tube.

16. A pole assembly, wherein the pole assembly comprises: a hollow inner cavity; an inner shell wall, wherein the inner shell wall comprises a first column; wherein the inner shell wall surrounds the hollow inner cavity; wherein the inner shell wall comprises steel; a center fill layer, wherein the center fill layer surrounds the inner shell wall; wherein the center fill layer comprises SSC concrete; and an outer shell wall, wherein the outer shell wall comprises a second column; wherein the outer shell wall comprises fiberglass reinforced polymer; wherein the outer shell wall surrounds the inner shell wall and the center fill layer.

17. The pole assembly of claim 16, wherein the center fill layer comprises a plurality of reinforcement wires which are arranged vertically within the center fill layer; wherein at least one reinforcement wire comprises high strength steel.

18. The pole assembly of claim 17, wherein at least one reinforcement wire comprises fiberglass reinforced polymer; wherein the center fill layer is formed of polyurethane.

19. The pole assembly of claim 18, wherein the center fill layer is formed of a first material at a first lower portion and a second fill material at a second upper portion.

20. The pole assembly of claim 19, wherein the inner shell wall is formed as a first circular column, and the outer shell wall is formed as a second circular column surrounding the first circular column.

21. The pole assembly of claim 17, wherein the center fill layer comprises a first reinforcement wire comprising high strength steel at a first lower level; wherein the center fill layer comprises a second reinforcement wire comprising fiberglass reinforced plastic at a second upper level, wherein the center fill layer comprises SSC at the first lower portion; wherein the center fill layer comprises polyurethane at the second upper portion.

22. The pole assembly of claim 21, wherein the center fill layer comprises a plurality of laterally extending reinforcing wires which are vertically spaced within the center fill layer; wherein the laterally extending reinforcing wires are formed as hoops.

23. The pole assembly of claim 22, wherein the center fill layer comprises a plurality of interior ribs; wherein at least one interior rib comprises two interlocking pieces which are mechanically engaged to form a single rib; wherein at least a portion of the single rib is attached to the outer FRP outer tube.

24. A pole assembly, wherein the pole assembly comprises: a spun prestressed concrete pole; an FRP tapered tube; wherein the FRP tapered tube is integrated within the concrete pole during manufacture; and an outer surface of the prestressed concrete pole.
